

THE 3-D BESSEL PROCESS

1. Introduction. Start with $\tilde{W} = (W_1, W_2, W_3)$ a three-dimensional Brownian motion, and look at

$$R_t = \|z + \tilde{W}_t\| = \sqrt{(z_1 + W_1(t))^2 + (z_2 + W_2(t))^2 + (z_3 + W_3(t))^2}.$$

This is the

Bessel process: a one-dimensional diffusion on $I = (0, \infty)$ with

$$R_t = r + \int_0^t \frac{ds}{R_s} + \beta_t, \quad r = \|z\| > 0$$

where β is another one-dimensional Brownian motion.

Diffusion $\sigma(r) = 1$, drift $b(r) = \frac{1}{r}$, scale function

$\rho(r) = -\frac{1}{r}$. In particular, $1/R_t$ is a local martingale, and is strict: $\mathbb{E}\left(\frac{1}{R_t}\right) \sim \frac{c}{\sqrt{t}}$ as $t \rightarrow \infty$.

Also, with $T_y = 1^{st} t \geq 0$ s.t. $R_t = y$ and $a < r < b$, we have then

$$P(T_a < T_b) = \frac{\frac{1}{r} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}, \quad \text{and sending } b \rightarrow \infty \text{ we obtain}$$

$$P(T_a < \infty) = P(\tau_0 \leq a) = \frac{a}{r} : \text{uniform distribution}$$

$$\text{for } \tau_0 \triangleq \inf_{t \geq 0} R_t.$$

2. Pitman's Theorem: Recall the equivalence in law of the two vector processes

$$\left(M_t^B - B_t, M_t^B \right)_{t \geq 0} \quad \text{and} \quad \left(|B_t|, L_t^B \right)_{t \geq 0}, \quad \text{where}$$

$L_t^B \triangleq \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \text{meas} \{0 \leq s \leq t: |B_s| \leq \varepsilon\}$ is the local time at the origin and

$M_t^B \triangleq \max_{0 \leq s \leq t} B_s$ is the maximum by time t , of the one-dimensional Brownian motion B .

Pitman's Theorem is the equivalence in law of the processes $\left(2M_t^B - B_t, M_t^B \right)_{t \geq 0}$ and $\left(R_t, J_t \right)_{t \geq 0}$,

where R is Bessel process and $J_t \triangleq \inf_{s \geq t} R_s \leq R_t$ its "historical low" after time t .

In particular, J has the same law as M^B , and $2J - R$ as Brownian motion B .

Proof: Start with the Bessel process R and define

$X_t \triangleq 2J_t - R_t$ (to play the role of Brownian motion).

We have

$$X_t = 2J_t - R_t \leq J_t \leq J_s \quad \text{for } s \geq t$$

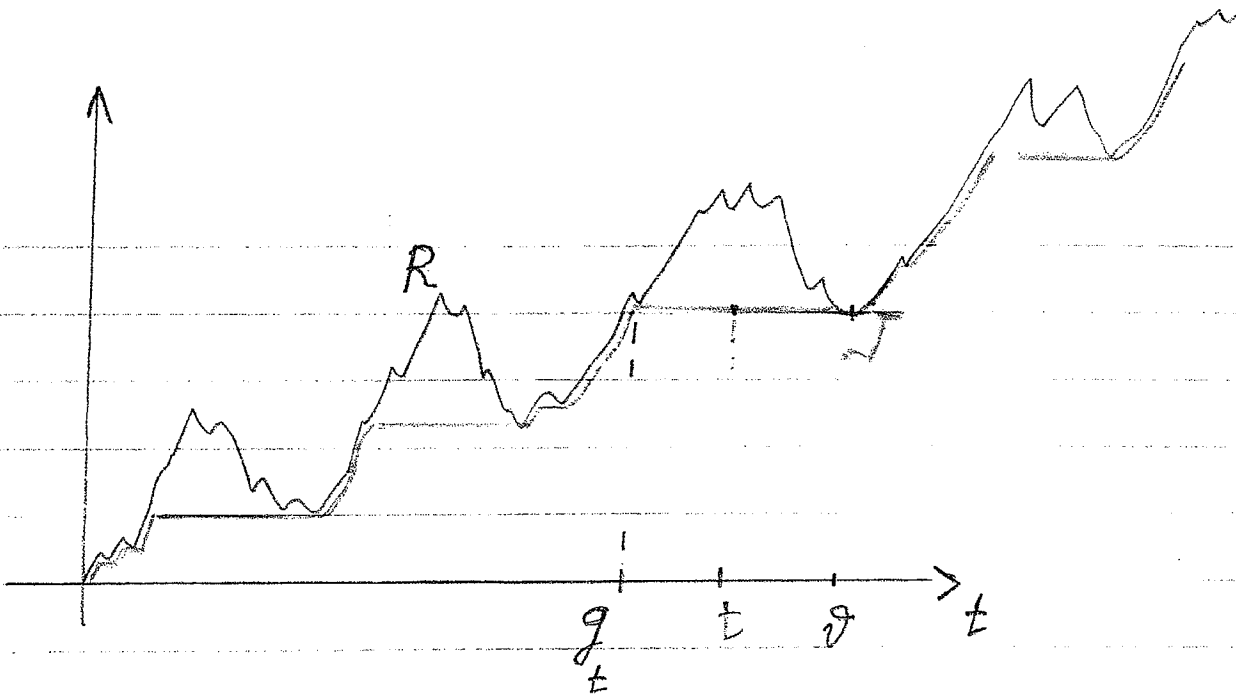
So $\sup_{s \leq t} X_s \leq J_t.$

We need to show that, in fact,

$$J_t = \sup_{s \leq t} X_s.$$

At any ϑ s.t. $J_\vartheta = R_\vartheta$,
we have

$$X_\vartheta = J_\vartheta, \quad R_s \geq J_s \text{ for } s < \vartheta$$



by definition, so $X_s = 2J_s - R_s \leq J_s \leq J_\vartheta = X_\vartheta$ holds for
all $s \leq \vartheta$ and

$$\text{thus: } J_\vartheta = \sup_{s \leq \vartheta} X_s.$$

On the other
hand, for any t s.t. $R_t > J_t$, look at

$$g = \sup \{s \leq t: J_s = R_s\}$$

so that

$$J_g = X_g, \quad X_t < J_t = J_g \text{ so again}$$

$$\sup_{s \leq t} X_s \geq X_t = J_t = J \geq \sup_{s \leq t} X_s \quad \square$$

Now we need to show that X is Brownian Motion. Clearly it is a continuous semimartingale of the filtration

$$\mathcal{G}_t \triangleq \mathcal{F}_t^{(R, J)} = \sigma(R_s, J_s; s \leq t)$$

with $\langle X \rangle_t = \langle R \rangle_t = t$. Thus, we need to show that it is a martingale with respect to its own filtration

$$\mathcal{F}_t^X = \sigma(X_s; s \leq t) \subseteq \mathcal{G}_t = \mathcal{F}_t^R \vee \sigma(J).$$

To justify this last claim, observe that knowledge of J_t and $(R_u, 0 \leq u \leq t)$ allows us to put together

$$J_s = \left(\inf_{s \leq u \leq t} R_u \right) \wedge J_t, \quad \text{"going backwards" from } t,$$

for all $0 \leq s \leq t$.

However, we also have

$$\sigma(J) \subseteq \mathcal{F}_t^J \subseteq \mathcal{F}_t^X$$

Since $J_t = \sup_{s \leq t} X_s$, and

$$R_t = 2J_t - X_t, \quad t \geq 0$$

so

$$\mathcal{F}_t^R \subseteq \mathcal{F}_t^X, \quad \text{so in fact}$$

$$\mathcal{F}_t^{2R-J} = \mathcal{F}_t^X = \mathcal{F}_t^{(R, J)} = \mathcal{F}_t^R \vee \sigma(J).$$

In other words, with $s < t$, in order to show

$$\mathbb{E}(X_t | \mathcal{F}_s^X) = X_s, \text{ a.s.}$$

it is enough to show

$$\mathbb{E}\left[X_t 1_{\{\tau > a\}} \mid \mathcal{F}_s^R\right] =: L =$$

$$\equiv K := \mathbb{E}\left[X_s 1_{\{\tau > a\}} \mid \mathcal{F}_s^R\right].$$

$\{R_s \leq a\}$, both K and L are $\equiv 0$.

• On the event $\{R_s > a\}$, the right-hand side is

$$K = \mathbb{E}\left[(2\tau - R_s) 1_{\{\tau > a\}} \mid R_s\right] \quad (\text{Markov property})$$

$$= \mathbb{E}\left[(2\tau - r) 1_{\{\tau > a\}} \mid R_s = r\right] \Big|_{r=R_s}$$

$$= \left[2 \int_a^r \frac{db}{r} - r \left(1 - \frac{a}{r}\right) \right] \Big|_{r=R_s}$$

$$= a - \frac{a^2}{R_s}, \text{ using the Markov property of } R.$$

Using this computation, we can write the left-hand side as

$$L = \mathbb{E}\left(X_t 1_{\{\tau > a\}} 1_{\{\tau_{s,t} > a\}} \mid \mathcal{F}_s^R\right)$$

$$= E \left[1_{\{J_{s,t} > a\}} \cdot E \left(X_t 1_{\{J_t > a\}} \mid \mathcal{F}_t^R \right) \mid \mathcal{F}_s^R \right]$$

$$= E \left[\left(a - \frac{a^2}{R_t} \right) \cdot 1_{\{J_{s,t} > a\}} \mid \mathcal{F}_s^R \right]$$

with $J_{s,t} \triangleq \inf_{s \leq u \leq t} R_u$. Again using the Markov property of R , we see that on the event $\{R_s > a\}$ this quantity is equal to

$$L = E \left[\left(a - \frac{a^2}{R_t} \right) 1_{\{\inf_{s \leq u \leq t} R_u > a\}} \mid R_s \right]$$

$$= E \left[\left(a - \frac{a^2}{R_{t-s}} \right) 1_{\{T_a > t-s\}} \mid R_0 = r \right] \Big|_{r=R_s}$$

$$= E \left(\left(a - \frac{a^2}{R_{(t-s) \wedge T_a}} \right) \mid R_0 = r \right) \Big|_{r=R_s}$$

$$= \left(a - \frac{a^2}{r} \right) \Big|_{r=R_s},$$

because $\frac{1}{R_{\cdot \wedge T_a}}$ is a bounded martingale.

□

Remark: From the identification $(M^B - B, M^B) = (|W|, L^W)$ we get

$$M_t^B = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \text{meas} \{0 \leq s \leq t : |W_s| \leq \varepsilon\} \\ = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \text{meas} \{0 \leq s \leq t : M_s^B - B_s \leq \varepsilon\}$$

is $\mathcal{F}_t^{M^B - B}$ - measurable. But then

$$B_t = M_t^B - (M_t^B - B_t)$$

is seen to be also

measurable, so that $\mathcal{F}_t^B \subseteq \mathcal{F}_t^{M^B - B}$. The reverse inclusion is trivial, so we get

$$\mathcal{F}_t^{M^B - B} = \mathcal{F}_t^B.$$

In contrast, the inclusion

$$\mathcal{F}_t^{2M^B - B} \subset \mathcal{F}_t^B \text{ is strict.}$$

In particular, the Bessel process $R = \sqrt{2M^B - B}$ is Markov process with respect to its own filtration $(\mathcal{F}_t^R)_{t \geq 0}$, but not with respect to $(\mathcal{F}_t^B)_{t \geq 0}$.

Indeed, the inclusion $\mathcal{F}_t^{2M^B - B} \subseteq \mathcal{F}_t^B$ holds trivially; suppose it is not strict, that is, $\mathcal{F}_t^B \subseteq \mathcal{F}_t^{2M^B - B}$ holds as well. Then there

would exist a non-anticipative functional $(t, x) \mapsto F_t^B(x)$ mapping $[0, \infty) \times C([0, \infty))$ into $\mathbb{R}$, such that

$$B_t^B = F_t^B(2M_t^B - B) \quad \text{for all } t \geq 0,$$

$$\text{and a fortiori: } 2M_t^B = F_t^B(2M_t^B - B) + (2M_t^B - B).$$

In other words, M_t^B would be completely determined, given $\mathcal{F}_t^{2M^B - B}$. That this is not the case is a direct consequence of the following result.

Corollary: The conditional distribution of M_t^B , given $\mathcal{F}_t^{2M^B - B}$, is uniform on $[0, 2M_t^B - B_t]$.

Proof: Recalling the identifications $M_t^B = J_t^B$ and $2M_t^B - B_t = R_t$ and the Markov property of R , we have

$$\begin{aligned} P[J_t^B < a \mid \mathcal{F}_t^R] &= P[\min_{s \geq t} R_s < a \mid R_t] \\ &= P[\min_{s \geq 0} R_s < a \mid R_0 = r] \Big|_{r=R_t} = \frac{a}{R_t}. \end{aligned}$$

Corollary: If R is a Bessel process with $R_0 = r > 0$ and $J_t^B = \min_{s \geq t} R_s$, then $(X_t = 2J_t^B - R_t, t \geq 0)$

is Brownian motion started at $2J_0 - r$, and
 $(X_t + r - 2J_0)_{t \geq 0}$ is standard Brownian motion and
 independent of $\mathcal{F}_0^X = \sigma(J_0)$.

Corollary: If W is Brownian motion and L^W its local
 time at the origin, then $|W| + L^W$ is Bessel.

Indeed, we have the P. Lévy identification
 $(|W|, L^W) = (M^B - B, M^B)$ for some other Brownian
 motion B , and so by Pitman's theorem

$$R \triangleq 2M^B - B = (M^B - B) + M^B = |W| + L^W$$

is a Bessel process.

Incidentally, recalling the
 Tanaka formula

$$|W| = \int_0^\cdot \operatorname{sgn} W_s \cdot dW_s + L^W$$

we deduce the
 decomposition of

$$R_t = \int_0^t \operatorname{sgn} W_s \cdot dW_s + 2L_t^W, \quad t \geq 0$$

as an $\{\tilde{F}_t^W\}$ - semimartingale.

Now R has a very different decomposition with respect to its own filtration, namely

$$R_t = r + \beta_t + \int_0^t \frac{ds}{R_s}, \quad t \geq 0$$

where β is an $(\mathcal{F}_t^R)$ -Brownian motion.

*) Proposition: Suppose that T is a stopping time of the filtration $\mathcal{G} = (\mathcal{F}_t^{R, J}) = (\mathcal{F}_t^B) = (\mathcal{F}_t^R \vee \sigma(J))_{t \geq 0}$, such that $R_T = J_T$. Then $(R_{T+t} - R_T)_{t \geq 0}$ is a Bessel process, independent of $(R_s, 0 \leq s \leq T)$.

Proof: Since B is Brownian motion, clearly $\tilde{B}_t = B_{T+t} - B_T$, $t \geq 0$ is standard Brownian motion, independent of $\mathcal{G}_T = \mathcal{F}_T^B$. Also $B_T = 2J_T - R_T = R_T = J_T$ and $J_{T+t} - J_T = \sup_{s \leq T+t} B_s - B_T =: M_t^{\tilde{B}}$, so that:

$$\begin{aligned} R_{T+t} - R_T &= 2J_{T+t} - B_{T+t} - R_T = 2(J_{T+t} - J_T) - (B_{T+t} - B_T) \\ &= 2M_t^{\tilde{B}} - \tilde{B}_t = \text{Bessel process by} \end{aligned}$$

Pitman's theorem, and independent of $\mathcal{F}_T^B$, thus also of $(R_s, 0 \leq s \leq T)$.

Proposition: Let R be Bessel process and B be Brownian motion with $R_0 = B_0 = 0$. For a given $b > 0$ let

$$L_b \triangleq \sup \{t \geq 0: R_t = b\}, \quad T_b \triangleq \inf \{t \geq 0: B_t = b\}.$$

Then

$$L_b \stackrel{d}{=} T_b.$$

Proof: We know we can take $R = 2M^B - B$, and of course

$$2M_{T_b}^B = 2b = b + B_{T_b}, \text{ so } T_b \leq L_b. \text{ On the other hand,}$$

$$\left(\overset{B_t}{B_{T_b+t}} \right)_{t \geq 0} = \left(B_{T_b+t} - B_{T_b} \right)_{t \geq 0} \text{ is standard Brownian motion, and thus}$$

$$M_{T_b+t}^B - M_{T_b}^B = M_t^{\tilde{B}} > 0 \text{ holds a.s.}$$

for all $t > 0$. Therefore

$$2M_{T_b+t}^B = M_{T_b+t}^B + M_{T_b+t}^B > b + B_{T_b+t}$$

for all $t > 0$, or equivalently

$$\text{But this gives } L_b \leq T_b. \quad \square$$

$$R_{T_b+t} > b.$$

In fact the following stronger result holds.

THEOREM: Let R be a Bessel process with $R_0 = 0$, and B be a Brownian motion with $B_0 = b > 0$. Define

$$L_b = \sup \{t \geq 0: R_t = b\}, \quad T_b = \inf \{t \geq 0: B_t = 0\}$$

Then

$$\left\{ R_{L_b-s}, 0 \leq s \leq L_b \right\} \stackrel{d}{=} \left\{ B_t, 0 \leq t \leq T_b \right\}.$$

3. D. Williams's Path Decomposition.

A G -stopping time that satisfies the properties of the previous Proposition, is the time (p.10)

$$\tau = \inf \{t \geq 0 : R_t = J_0\}$$

at which the Bessel process R attains its all-time minimum $J_0 = \inf_{s \geq 0} R_s$. After this time the process

$(R_{\tau+t} - R_\tau)_{t \geq 0}$ is again Bessel, so R never again visits J_0 : this τ is the unique instant at which $R_\tau = J_0$.

We also know that

$$R_\tau = J_0 \text{ is uniform on } [0, r].$$

Proposition: Let R be a Bessel process with $R_0 = r > 0$.

Then $\{R_t, 0 \leq t < \tau\}$ is equivalent in law to

$\{B_t, 0 \leq t < T_Y\}$, where B is Brownian motion with $B_0 = r$ and $T_Y = \inf \{t \geq 0 : B_t = Y\}$ with Y independent of (B, R) and uniformly distributed on $[0, r]$.

Proof: We already know (Corollary on p.8) that

$$B_t = 2J_t - R_t, t \geq 0 \text{ is Brownian motion with } B_0 = 2J_0 - r.$$

For $t < \tau$ we have $J_t = J_0$, so $B_t = 2J_t - R_t$ and $R_t = 2J_0 - B_t =: \beta_t$, a Brownian motion with $\beta_0 = r$.

Now again for $t < \tau$ note

$\beta_t = \beta_0 - (B_t - B_0) = r - (B_t - B_0)$ is independent of B_0 , thus

R_t is independent of $J_0 = R_\tau$ and $\tau = \inf\{t \geq 0: \beta_t = J_0\}$.

Because J_0 is uniformly distributed on $[0, r]$, we are done.

D. Williams: Path Decomposition. Pick a number $c > 0$ and the following four independent elements:

- Two Bessel processes $R, \tilde{R}$ with $R_0 = \tilde{R}_0 = 0$;
- A random variable ξ uniformly distributed on $[0, c]$;
- A Brownian motion B with $B_0 = c$.

Define $S_c = \inf\{t \geq 0: R_t = c\}$, $T_\xi = \inf\{t \geq 0: B_t = \xi\}$.

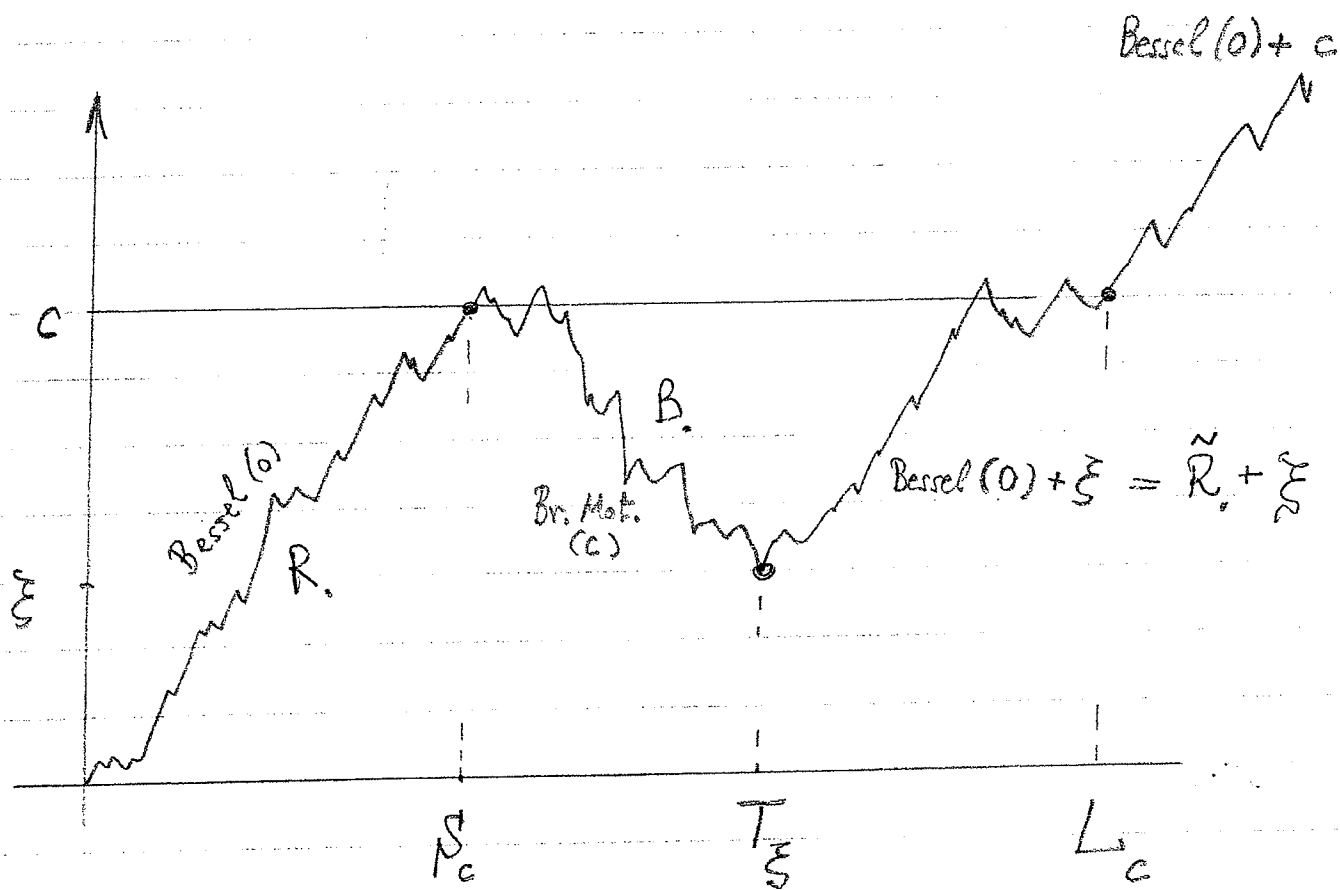
Then

$$Z_t \triangleq \begin{cases} R_t & \text{for } 0 \leq t < S_c \\ B_{t-S_c} & \text{for } S_c \leq t < S_c + T_\xi \\ \xi + \tilde{R}_{t-(S_c+T_\xi)} & \text{for } t \geq S_c + T_\xi \end{cases}$$

is Bessel process with

$$Z_0 = 0.$$

Proof: If we look at a Bessel process $(p_t)_{t \geq 0}$ with $p_0 = 0$, then the strong Markov property gives that $\{p_t, t < S_c\}$ and $\{p_{t+S_c}, t \geq 0\}$ are independent, and the second one is Bessel process starting at c . Thus the theorem follows from the preceding results.



Remark:

$$L_c = \sup \{ t \geq 0 : R_t = c \}$$

$$= \inf \{ t \geq 0 : R_t = \underline{J}_t = c \}$$

so L_c is as in Proposition, p. 10.

Corollary: If R is Bessel process with $R_0 = r > 0$, then $\tilde{B} \triangleq 1/R$ is time-changed Brownian motion on $[0, T_0)$ with $\tilde{T}_0 = \inf\{t \geq 0 : \tilde{B}_t = 0\}$.

Indeed, we know that $R_t = r + \int_0^t \frac{ds}{R_s} + \beta_t$, $t \geq 0$ where β is standard Brownian motion, so Ito's rule gives $\tilde{B}_t = \frac{1}{r} - \int_0^t \frac{d\beta_s}{R_s^2} = \frac{1}{r} + \int_0^t A_s$ where $\int$ is standard Brownian motion and $A_t \triangleq \int_0^t \frac{ds}{R_s^4}$. But we know that $\lim_{t \rightarrow \infty} R_t = \infty$ a.s. so $\lim_{t \rightarrow \infty} \tilde{B}_t = 0$ a.s.

and this means: $A_\infty = \tilde{T}_0$.

PROPOSITION: Suppose that B is Brownian motion with $B_0 = a > 0$, and define $T_0 = \inf\{t \geq 0 : B_t = 0\}$,

$$M = \sup_{0 \leq t < T_0} B_t.$$

Then we have the following:

$$(i) \quad P[M \geq b] = \frac{a}{b}, \text{ for } b \geq a.$$

(ii) There exists a unique time $\nu \in (0, T_0)$ st. $B_\nu = M$.

Furthermore, conditionally on the event $\{M = m\}$, the processes

(iii) $X^1 = \{B_t, t < \nu\}$ and $X^2 = \{B_{\nu+u}, 0 \leq u < T_0 - \nu\}$ are independent,

(iv) X^1 is Bessel, run until it hits m ,

(v) $m - X^2$ is Bessel, run until it hits m .

Proof of (i), (ii): From the Corollary we know that $\{B_s, 0 \leq s < T_0\}$ has the same distribution as

$\left\{ \frac{1}{R_{C_s}}, 0 \leq s < A_\infty \right\}$, where R is Bessel process with $R_0 = r = \frac{1}{a}$ and C is the inverse of $A = \int_0^\cdot \frac{ds}{R_s^4}$.

Therefore, $M = \frac{1}{\inf_{0 \leq t < \infty} R_t} = \frac{1}{\inf_{0 \leq t < \infty} R_t}$, and from our discussion on pp. 11

we know $M = B_\nu$ for exactly one ν in $(0, T_0)$, and

$$P(M \geq b) = P\left(\inf_{0 \leq t < \infty} R_t \leq \frac{1}{b}\right) = \frac{1/b}{r} = \frac{a}{b}.$$

Thus, we shall

identify A_∞ with T_0 , and B with $1/R_C$, $\frac{1}{R}$ with $\frac{B}{A}$.

Proof of (iii): Recall $\tau \triangleq \inf\{t \geq 0 : R_t = \sqrt{0}\} = C(\nu)$ where

$$\nu \triangleq \inf\{s < T_0 : B_s = M\} = \inf\{s < A_\infty : R_{C_s} = \frac{1}{M}\}$$

We have $R_\tau = \sqrt{0} = \frac{1}{M} = \sqrt{\nu}$,

and so from Prop. on p. 10:

$\{R_{\tau+t} - R_\tau, t \geq 0\}$ is Bessel, independent of $\{R_t, 0 \leq t \leq \tau\}$.

Rephrasing, we get the result.

Proof of (iv): We start by observing

$$(X^1 | M=m) = \left(\frac{1}{R_{C_s}}, s < A_\tau \mid R_\tau = \frac{1}{m} \right).$$

But Proposition on p. 11 says that there exists a Brownian motion $\tilde{X}$ with $\tilde{X}_0 = \frac{1}{a}$, such that

$$\left(\frac{1}{R_t}, t < \tau \mid R_\tau = \frac{1}{m} \right) \stackrel{L}{=} \left(\frac{1}{\tilde{X}_t}, t < T_{1/m} \right)$$

and consequently where $(X^1 / M=m) \stackrel{L}{=} \left(\frac{1}{\xi_{C_s}}, s < A_{T_{1/m}} \right)$

$$A_t \triangleq \int_0^t \frac{du}{\xi_u^4} \quad \text{and} \quad A_{C_s} = s. \quad \text{In particular,}$$

$$\left. \frac{dA_t}{dt} \right|_{t=C_s} \cdot \frac{dC_s}{ds} = 1, \quad \text{so} \quad \frac{dC_s}{ds} = \xi_{C_s}^4.$$

But from Ito's rule and DDS we know

$$\frac{1}{\xi_t} = a - \int_0^t \frac{d\xi_u}{\xi_u^2} + \int_0^t \frac{du}{\xi_u^3} = a + \beta_{A_t} + \int_0^t \frac{du}{\xi_u^3}$$

where β is Brownian motion, so that

$$\frac{1}{\xi_{C_s}} = a + \beta_s + \int_0^s \frac{dC_v}{\xi_{C_v}^3} = a + \beta_s + \int_0^s \xi_{C_v} dv.$$

In other words, $\frac{1}{\xi_{C_s}}$ satisfies the s.d.e. of a Bessel process as claimed.